

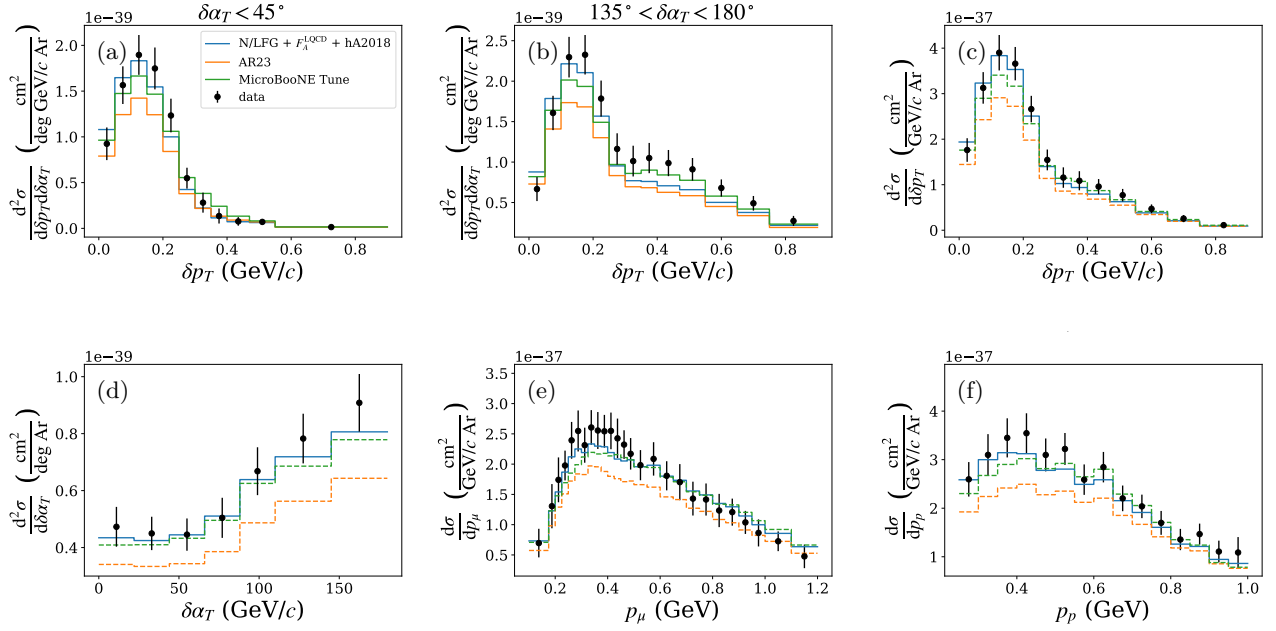


Figure 1: The fully theory-driven GENIE configuration $N/LFG + F_A^{\text{LQCD}} + hA2018$ (blue) compared with the AR23 (orange) and MicroBooNE (green) reference tunes on the MicroBooNE $\text{CC}1\mu1p / \text{CC}1\mu\text{Np}$ argon cross sections (black points). Panels: (a) δp_T for $\delta\alpha_T < 45^\circ$ and (b) for $135^\circ < \delta\alpha_T < 180^\circ$ (two slices of the 2D δp_T – $\delta\alpha_T$ measurement); (c) δp_T ; (d) $\delta\alpha_T$; (e) p_μ ; (f) p_p . The reference tunes are drawn dashed in the projected panels (c)–(f). The $N/LFG + F_A^{\text{LQCD}} + hA2018$ prediction gives the lowest χ^2/ndf of the three in every observable (see Table 1).

Table 1: χ^2/ndf (p -value) relative to MicroBooNE data for Fig. 1.

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
$N/LFG + F_A^{\text{LQCD}} + hA2018$	35.98/49 (0.92)	6.78/13 (0.91)	3.47/7 (0.84)	16.29/26 (0.93)	16.05/15 (0.38)
AR23	38.05/49 (0.87)	10.63/13 (0.64)	7.56/7 (0.37)	28.96/26 (0.31)	24.07/15 (0.06)
MicroBooNE Tune	42.61/49 (0.73)	7.31/13 (0.89)	3.96/7 (0.78)	24.59/26 (0.54)	17.38/15 (0.30)